

1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Experiment 1.1

% Generation of Continuous Time Signals

clc;

clear;

close all;

% Time axis

t = -5:0.01:5;

% Continuous-time sine signal

x = sin(2*pi*t);

% Plotting

figure;

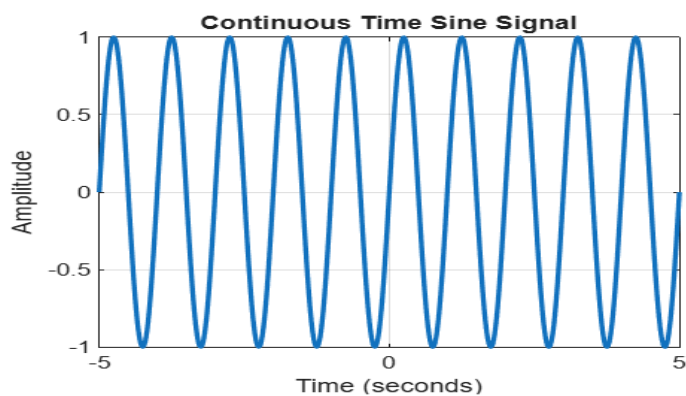
plot(t, x, 'LineWidth', 2);

grid on;

title('Continuous Time Sine Signal');

xlabel('Time (seconds)');

ylabel('Amplitude');



Experiment 1.2

% Generation of Discrete Time Signal

clc;

clear;

close all;

% Discrete-time index

n = -10:10;

% Discrete-time sinusoidal signal

x = sin(0.2*pi*n);

% Plotting

figure;

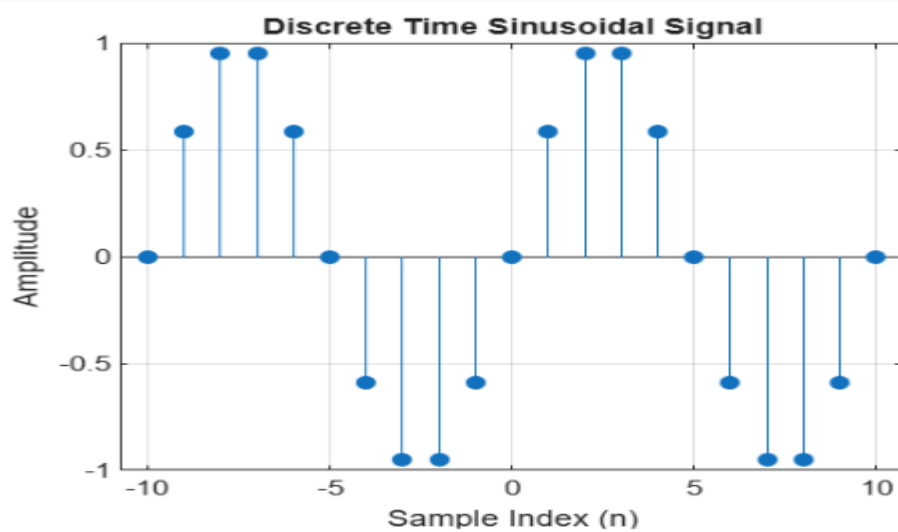
stem(n, x, 'filled');

grid on;

title('Discrete Time Sinusoidal Signal');

xlabel('Sample Index (n)');

ylabel('Amplitude');



Experiment 1.3

% Generation of Triangular and Sawtooth Signals

clc;

clear;

close all;

% Time axis

t = 0:0.01:2;

% Sawtooth Signal

x1 = sawtooth(2*pi*t);

% Triangular Signal

x2 = sawtooth(2*pi*t,0.5);

% Plotting

figure;

subplot(2,1,1);

plot(t,x1,'LineWidth',2);

grid on;

title('Sawtooth Signal');

xlabel('Time (s)');

ylabel('Amplitude');

subplot(2,1,2);

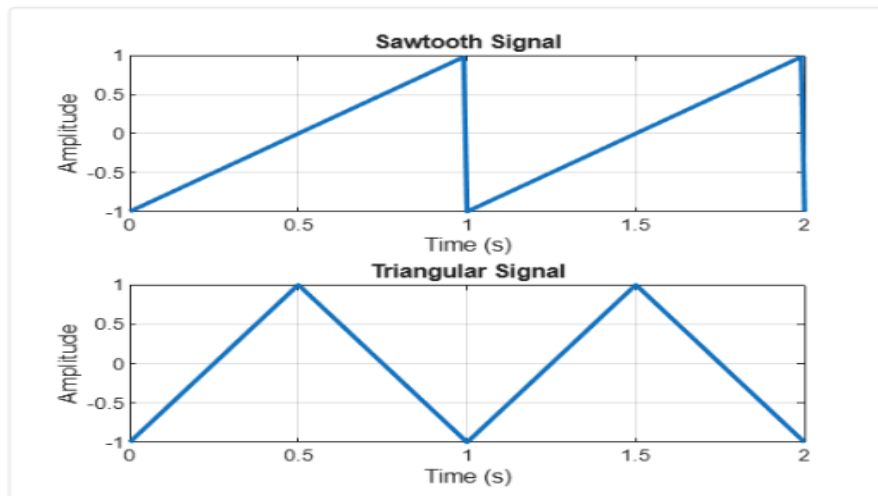
plot(t,x2,'LineWidth',2);

grid on;

title('Triangular Signal');

xlabel('Time (s)');

```
ylabel('Amplitude');
```



Experiment 1.4

% Step and Ramp Functions (Continuous & Discrete)

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Continuous-Time Signals
```

```
t = -5:0.01:5;
```

```
% Continuous Unit Step Function
```

```
u = double(t >= 0);
```

```
% Continuous Ramp Function
```

```
r = t .* (t >= 0);
```

```
%% Discrete-Time Signals
```

```
n = -5:5;
```

```
% Discrete Unit Step Function
```

```
ud = double(n >= 0);  
% Discrete Ramp Function  
rd = n .* (n >= 0);  
%% Plotting  
figure;  
subplot(2,2,1);  
plot(t,u,'LineWidth',2);  
grid on;  
title('Continuous Unit Step Function');  
xlabel('Time (t)');  
ylabel('Amplitude');  
subplot(2,2,2);  
plot(t,r,'LineWidth',2);  
grid on;  
title('Continuous Ramp Function');  
xlabel('Time (t)');  
ylabel('Amplitude');  
subplot(2,2,3);  
stem(n,ud,'filled');  
grid on;  
title('Discrete Unit Step Function');  
xlabel('Sample Index (n)');  
ylabel('Amplitude');  
subplot(2,2,4);
```

```
stem(n,rd,'filled');
```

```
grid on;
```

```
title('Discrete Ramp Function');
```

```
xlabel('Sample Index (n)');
```

```
ylabel('Amplitude');
```

